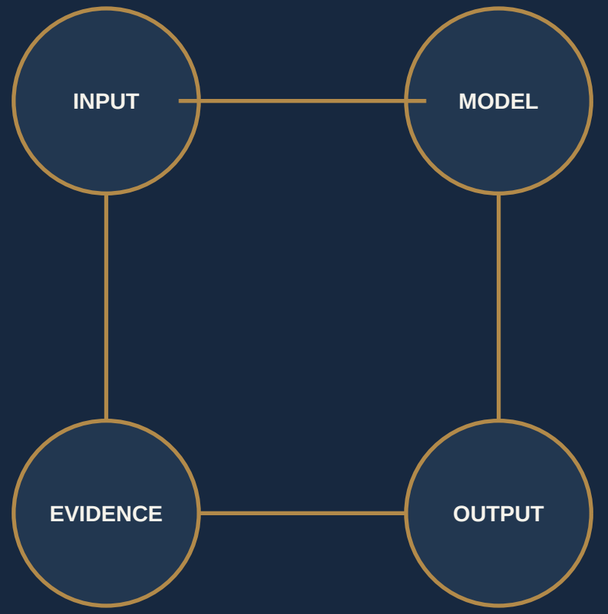
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Human-Robot Interaction

AI.110 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define interaction design.
- Apply key methods including human perception.
- Build or analyze a practical system involving speech interaction.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Interaction design

02

METHODS

Human perception

03

SYSTEMS

Speech interaction

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Interaction design

Interaction design

02 · Human perception

Human perception

03 · Speech interaction

Speech interaction

04 · Gestures

Gestures

05 · Social cues

Social cues

06 · Trust

Trust

07 · Robot interfaces

Robot interfaces

08 · Collaborative robots and human

Collaborative robots and human factors

Core ideas, made explicit

AI.110 • Distinguish definitions, assumptions and evidence

- Interaction design
- Human perception
- Speech interaction
- Gestures
- Social cues

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.110 • Distinguish definitions, assumptions and evidence

- Trust
- Robot interfaces
- Collaborative robots and human factors
- Realistic human-robot interaction footage and simulations
- Mandatory practicals: Design and test a human-robot interface

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM
Interaction design

INPUTS
Human perception

METHOD
Speech interaction

OUTPUT
Gestures

EVALUATION
Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving interaction design.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using interaction design.
- Implement or simulate a baseline using human perception.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.